

■ Python Theory-Based MCQs

1. What is the correct file extension for Python files?

- A. .pt
- B. .pyt
- C. .py
- D. .pyn

Answer: C. .py

2. What does the `len()` function do?

- A. Returns the memory address
- B. Returns the number of items in an object
- C. Deletes an object
- D. Converts string to integer

Answer: B. Returns the number of items in an object

3. What is the output of `type([])`?

- A. <class 'list'>
- B. <type 'list'>
- C. list
- D. <list>

Answer: A. <class 'list'>

4. Which of the following is not a valid keyword in Python?

- A. pass
- B. eval
- C. assert
- D. nonlocal

Answer: B. eval (It's a built-in function, not a keyword)

5. Which operator is used for exponentiation in Python?

- A. ^
- B. **
- C. exp()
- D. %

Answer: B. **

Output-Based Python MCQs

6. What is the output of the following code?

```
x = 5
print(x == 5.0)
```

- A. True
- B. False
- C. Error
- D. None

Answer: A. True (Because `5 == 5.0` is True in Python)

7. What is the output of the following code?

```
a = "Python"
print(a[1:4])
```

- A. yth
- B. Pyt
- C. ytho
- D. ytho

Answer: A. yth

8. What is the result of this code?

```
print(2 ** 3 ** 2)
```

- A. 64
- B. 512
- C. 256
- D. 36

Answer: B. 512

Explanation: `3 ** 2 = 9`, then `2 ** 9 = 512`

9. What will be printed by this code?

```
print(bool(0), bool(3.14), bool(-3))
```

- A. False False False
- B. True True True
- C. False True True
- D. Error

Answer: C. False True True

10. What is the output of this code?

```
x = "Hello"  
print(x * 3)
```

- A. HelloHelloHello
- B. Hello3
- C. Error
- D. Hello Hello Hello

Answer: A. HelloHelloHello

11. What is the output of `type((1))` in Python?

- A. <class 'tuple'>
- B. <class 'int'>
- C. <class 'list'>
- D. <class 'str'>

Answer: B. <class 'int'>

12. What will be the type of variable `x = 10.0`?

- A. int
- B. float
- C. double
- D. long

Answer: B. float

13. Which of the following is mutable?

- A. str
- B. tuple
- C. int
- D. list

Answer: D. list

14. What keyword is used to define a function in Python?

- A. define
- B. function
- C. def
- D. fun

Answer: C. def

15. How do you insert COMMENTS in Python code?

- A. `// comment`
- B. `<!-- comment -->`
- C. `# comment`
- D. `/* comment */`

Answer: C. `# comment`

16. What does the `range(5)` function return?

- A. A list from 1 to 5
- B. A list from 0 to 5
- C. A range object with numbers 0 to 4
- D. Error

Answer: C. A range object with numbers 0 to 4

17. Which function is used to get user input in Python 3?

- A. `raw_input()`
- B. `get()`
- C. `input()`
- D. `scan()`

Answer: C. `input()`

18. What is the output type of `input()` in Python 3?

- A. `str`
- B. `int`
- C. depends on input
- D. `bool`

Answer: A. `str`

19. Which of the following is used for integer division in Python?

- A. `/`
- B. `//`
- C. `%`
- D. `div`

Answer: B. `//`

20. What is the keyword to start a loop in Python?

- A. `repeat`
- B. `for`
- C. `foreach`

D. loop

Answer: B. for

21. Which of the following is not a built-in data type in Python?

A. list

B. dict

C. array

D. set

Answer: C. array

22. What is the correct way to define a dictionary?

A. {}

B. []

C. ()

D. <>

Answer: A. {}

23. What does the `break` statement do in a loop?

A. Skips one iteration

B. Exits the loop

C. Restarts the loop

D. Exits program

Answer: B. Exits the loop

24. How do you start a conditional block in Python?

A. `if x then:`

B. `if x:`

C. `if(x)`

D. `if then x:`

Answer: B. `if x:`

25. Which keyword is used to handle exceptions in Python?

A. exception

B. try

C. error

D. catch

Answer: B. try

26. What is the output?

```
print("4" + "2")
```

- A. 6
- B. 42
- C. Error
- D. None

Answer: B. 42

27. What is the output?

```
print(10 / 3)
```

- A. 3
- B. 3.0
- C. 3.3333333333333335
- D. 3.33

Answer: C. 3.3333333333333335

28. What is the output?

```
print(10 // 3)
```

- A. 3
- B. 3.0
- C. 3.333
- D. Error

Answer: A. 3

29. What is the output?

```
print("Python"[::-1])
```

- A. Python
- B. nohtyP
- C. Error
- D. nothyP

Answer: B. nohtyP

30. What is the output?

```
print(len("Python"))
```

- A. 5
- B. 6
- C. 7
- D. Error

Answer: B. 6

31. What is the output?

```
print([1, 2] + [3, 4])
```

- A. [1, 2, 3, 4]
- B. 10
- C. Error
- D. [1, 2][3, 4]

Answer: A. [1, 2, 3, 4]

32. What is the output?

```
x = [1, 2, 3]  
print(x * 2)
```

- A. [1, 2, 3, 1, 2, 3]
- B. [2, 4, 6]
- C. Error
- D. [1, 2, 3, 2]

Answer: A. [1, 2, 3, 1, 2, 3]

33. What is the output?

```
print(type({1, 2, 3}))
```

- A. <class 'list'>
- B. <class 'dict'>
- C. <class 'set'>
- D. <class 'tuple'>

Answer: C. <class 'set'>

34. What is the output?

```
print(bool(""))
```

- A. True
- B. False
- C. None

D. Error

Answer: B. False

35. What is the output?

```
print(5 > 3 and 2 < 1)
```

A. True

B. False

C. Error

D. None

Answer: B. False

36. What is the output?

```
print(5 > 3 or 2 < 1)
```

A. True

B. False

C. None

D. Error

Answer: A. True

37. What is the output?

```
x = 5
x += 3
print(x)
```

A. 8

B. 5

C. 3

D. Error

Answer: A. 8

38. What is the output?

```
for i in range(3):
    print(i, end=" ")
```

A. 0 1 2

B. 0 1 2

C. 012

D. 0, 1, 2

Answer: B. 0 1 2

39. What is the output?

```
a = [1, 2, 3]
a.append(4)
print(a)
```

A. [1, 2, 3]

B. [1, 2, 3, 4]

C. Error

D. None

Answer: B. [1, 2, 3, 4]

40. What is the output?

```
a = [1, 2, 3]
a.pop()
print(a)
```

A. [1, 2, 3]

B. [1, 2]

C. [1]

D. []

Answer: B. [1, 2]

41. What is the output?

```
print(bool(None))
```

A. True

B. False

C. None

D. Error

Answer: B. False

42. What is the output?

```
a = (1, 2, 3)
print(a[1])
```

A. 1

B. 2

- C. 3
- D. Error

Answer: B. 2

43. What is the output?

```
print(10 % 3)
```

- A. 3
- B. 1
- C. 0
- D. 10

Answer: B. 1

44. What is the output?

```
a = {"key": "value"}  
print(a["key"])
```

- A. key
- B. value
- C. Error
- D. None

Answer: B. value

45. What is the output?

```
print("A" > "a")
```

- A. True
- B. False
- C. Error
- D. Depends

Answer: B. False

46. What is the output?

```
print([i for i in range(3)])
```

- A. [0, 1, 2]
- B. range(3)
- C. 012
- D. Error

Answer: A. [0, 1, 2]

47. What is the output?

```
x = None  
print(x is None)
```

- A. True
- B. False
- C. Error
- D. None

Answer: A. True

48. What is the output?

```
print(0 and 1)
```

- A. 1
- B. 0
- C. True
- D. False

Answer: B. 0

49. What is the output?

```
print(1 or 0)
```

- A. 1
- B. 0
- C. True
- D. False

Answer: A. 1

50. What is the output?

```
print(not 0)
```

- A. 0
- B. 1
- C. True
- D. False

Answer: C. True

1. What is the output of the following code?

```
print(type("Hello"))
```

- A. <class 'str'>
- B. <class 'char'>
- C. <class 'text'>
- D. str

Answer: A. <class 'str'>

2. Which of the following is used to define a block of code in Python?

- A. Curly braces {}
- B. Indentation
- C. Parentheses ()
- D. Colon :

Answer: B. Indentation

3. What will be the output of this code?

```
x = [1, 2, 3]
x[0] = 10
print(x)
```

- A. [10, 2, 3]
- B. [1, 2, 3]
- C. [10, 1, 2, 3]
- D. Error

Answer: A. [10, 2, 3]

4. Which of the following is not a valid Python data type?

- A. set
- B. list
- C. array
- D. tuple

Answer: C. array

5. What is the output of the following expression?

```
3 * 1 ** 3
```

- A. 27
- B. 3
- C. 1
- D. 9

Answer: B. 3

Explanation: $1 ** 3 = 1$, then $3 * 1 = 3$

6. Which function is used to find the largest item in an iterable?

- A. `top()`
- B. `large()`
- C. `max()`
- D. `big()`

Answer: C. `max()`

7. What will this code output?

```
print("2" * 3)
```

- A. 222
- B. 6
- C. Error
- D. 2 2 2

Answer: A. 222

8. Which symbol is used for comments in Python?

- A. `//`
- B. `<!-- -->`
- C. `#`
- D. `/* */`

Answer: C. `#`

9. Which method is used to add an element at the end of a list?

- A. `add()`
- B. `append()`
- C. `insert()`
- D. `push()`

Answer: B. `append()`

10. What is the output of this code?

```
x = "abc"
print(x.upper())
```

- A. abc
- B. ABC

C. Abc

D. error

Answer: B. ABC
